On the evaluation of StartupGPT: A Retrieval-Augmented AI Chatbot for Delivering Research-Driven Guidance to Startups

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***Abstract*—[Introduction] The study addresses the challenge of transferring research knowledge to the industry, with a particular focus on small businesses and startups, which often lack access to empirical insights. Large Language Models (LLMs), particularly those using Retrieval-Augmented Generation (RAG), offer potential for embedding knowledge from startup research into an interactive chatbot to support startup mentorship. However, empirical work exploring this application is limited. [Objective] The primary objective of this research is to design and evaluate a version of "StartupGPT," an AI-driven chatbot that uses LLMs and RAG to provide advice for software startups by leveraging a knowledge base rooted in software startup research. [Methodology] The study follows the Design Science Research Methodology (DSRM) and spans three iterative cycles, with this paper focusing on Cycle 3. The prototype was tested with 11 startup founders, who provided both qualitative and quantitative feedback on the chatbot's usefulness and satisfaction. [Results] The findings from user tests indicate that StartupGPT was generally perceived as relevant, reliable, and helpful. However, limitations were noted in its responses, which users found overly theoretical, lacking in concrete examples, and insufficiently personalized for specific startup contexts. [Conclusion] Future LLM-based interaction designed for startups should focus on improving interactivity, incorporating more context-aware and specific advice, and leveraging advanced AI techniques, such as fine-tuning, to better align the chatbot's responses with the unique needs of individual startups.**

***Index Terms*—Startup, large language models, Retrieval-Augmented Generation, Online Mentor, Startup Mentor.**

I. Introduction

The field of empirical software engineering aims to produce research outcomes that practitioners can utilize to make informed decisions in software development [1]. However, software engineering research often struggles to demonstrate its practical impact due to challenges like limited time commitment, ineffective reward systems, and perceived lack of utility in the industry [2]. Although the theoretical foundation of software engineering research remains under debate, empirical insights can provide valuable support to practitioners dealing with real-world problems. Bridging the gap between research and practice is an essential topic in fields like Information Systems and Software Engineering. In Information Systems, evidence-based practice "an approach to decision making in the design, adoption and implementation of information systems that uses the best evidence available, from both practitioner expertise and systematic research, in consultation with all stakeholders" [3]. A key barrier, however, is the focus of academic research on developing generalizable principles, while practitioners often operate in ad hoc contexts, dealing with situational factors unique to each project [4]. Additionally, most research on industry-academia collaboration centers around large companies, often overlooking the unique needs and constraints faced by smaller firms and startups.

Research on software startups, as a branch of empirical software engineering, similarly seeks to improve the practices of startup founders and developers [5]. Prior studies have highlighted the entrepreneurial, effectual logic that drives startup decision-making, emphasizing reliance on available resources, evolving goals, and adaptability to uncertain environments [6]. In such an effectual approach, entrepreneurs prioritize rapid, actionable information to inform flexible strategies. While software startups could significantly benefit from empirical evidence, best practices, and case studies, the absence of a convenient interface for interacting with and absorbing this knowledge creates a substantial barrier. Traditional sources of support, such as startup incubators and accelerators, have become prominent; however, access to these resources can be limited by geographical, financial, and capacity constraints.

The potential of generative AI, particularly large language models (LLMs), to support startups is promising [7]. LLMs can generate human-like responses to questions, potentially

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assisting companies in areas like operational efficiency and creative problem-solving [7]. Moreover, the adoption of LLM tools like ChatGPT and CoPilot is increasingly popular in software industry. However, a key limitation of LLMs is their tendency to produce plausible yet inaccurate or irrelevant outputs—commonly termed ”hallucinations.” This presents a significant challenge in the context of providing precise, reliable advice to early-stage entrepreneurs, where accuracy is critical. Addressing this limitation, Retrieval-Augmented Generation (RAG) offers a compelling solution. RAG enhances the accuracy of LLMs by integrating information retrieved in real-time from an external knowledge base, reducing the risk of hallucinations and improving response quality [8], [9]. This approach could enable LLMs to act as reliable, scalable, and accessible startup mentors, filling the gap left by limited access to traditional mentorship resources.

In this study, we investigate the following research question: How software startup research outcome can be integrated in LLMs to assist software startups in business and product development? By leveraging the knowledge extracted from software startup research via RAG technique, we developed a prototype of StartupGPT, a LLM-based virtual assistant for software startups. This empirical case study contributes to the field by addressing two primary gaps: the limited transfer of knowledge from academia to software startups and the underexplored potential of large language models (LLMs) to support startup-specific challenges.

The structure of this paper is as follows: Section 2 provides the design of StartupGPT, describing the integration of RAG and outlining the knowledge base development process. Section 3 presents the methodology for user testing. Section 4 presents the result. Section 5 discusses and conclude the paper.

II. Design of StartupGPT

StartupGPT is an LLM-based chatbot that assists startups in developing their business ideas. StartupGPT aims to provide a flexible solution that facilitates the testing and evaluation of various chatbot modifications to determine their effectiveness in addressing the specific challenges faced by startups in regard to idea validation. The tool consists of three chatbots, modified with carefully crafted system instructions based on prompt engineering techniques and **rag**. The system instructions, additionally, dynamically include demographic information about the entrepreneurs and their startup to enable targeted and specific outputs from the chatbots.

A. Architectural view of StartupGPT

The prototype was developed in the open-source Python framework Streamlit[1]. Streamlit is designed to simplify the creation of interactive and shareable web applications, and it offers the advantage of free deployment through its servers [10]. The platform’s capability to enable the rapid creation and easy sharing of the prototype to potential user testers through a web link made it a suitable choice for this research. All chatbots in StartupGPT were based on *GPT-4*

[1]https://streamlit.io/

Users can input their startup-related problems on the StartupGPT website. When users interact with the application, their requests are sent to the database through the Streamlit application server and OpenAI API. When the user asks a question to a chatbot, their question is sent to and processed by the OpenAI API, and its response is sent back to the application server and then the user. Information from the forms and chatlogs is saved in the database. An interface of StartupGPT for user testing is shown in Figure 1.

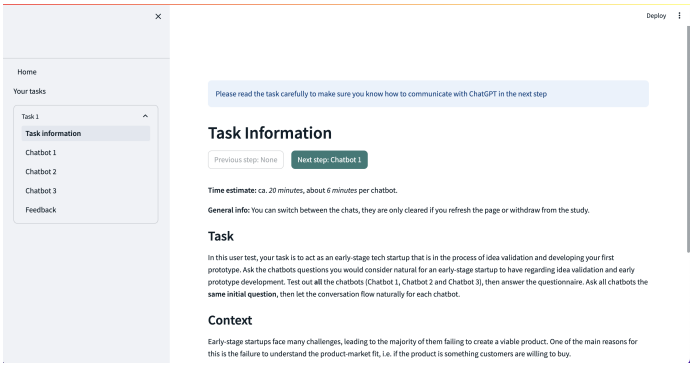


Fig. 1. Architecture of StartupGPT

The website outputs responses from the chatbots. After the reactions, the user provides feedback on the responses generated by the chatbots regarding their usefulness. The overall interaction data is stored for in-depth analysis.

B. Embedding empirical software startup research with RAG

A RAG model consists of two steps: embedding and retrieval. An illustration of the retrieval process of the documents can be seen in Figure 2. The users’ query is turned into a vector embedding. The user query is then compared with the vector embedding in the database using a cosine similarity function. The three information chunks with the highest similarity score are retrieved from the knowledge base and returned to the Streamlit application server. These information chunks are added to the system instruction, together with the user’s question, and used as context for the chatbot when generating an answer from the OpenAI.

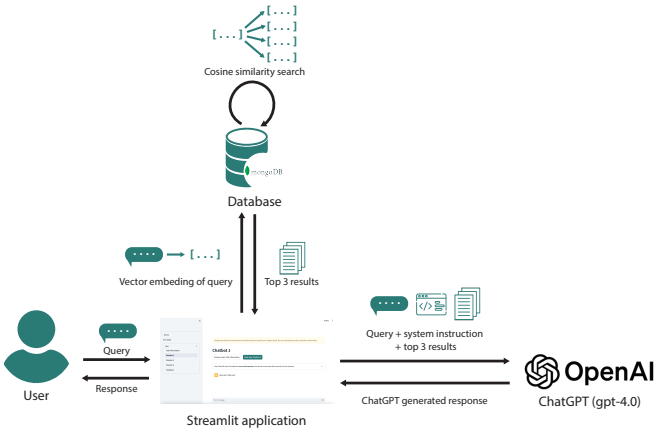
1) Knowledge Base of Documents: The knowledge base for the **RAG** model was created from 20 research articles. The number of materials is limited due to the budget to build RAG in this experiment. The criteria for selecting materials for RAG include:

1) Relevance to the domain: content should come from software startup research
2) Content quality: prioritize materials from high-quality sources (well-known journals, well-known authors or websites)
3) Robustness for extract: materials appear a structured text (PDF, structure website, etc)

A combination of peer-reviewed articles and books about software startup frameworks and blog posts by leading entrepreneurs in the field was used. This was to ensure both

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Fig. 2. Complete RAG pipeline for StartupGPT



academic and more practical literature. All documents were quality assured by the project supervisor, in addition to both researchers. For the implementation of the functionality to build the RAG model, components by LangChain were utilised.

2) Chunking the Documents: The documents used in the knowledge base of a **RAG** model are split into smaller pieces, called chunks. A recursive chunking was selected for this project to keep semantically related parts together. A chunk size of 2000 tokens, with a 200-token overlap, was selected through iterative testing and internal evaluation. For this study, larger chunks were considered favorable to ensure sufficient context. This consideration, in combination with a Microsoft guide[2] using the average and maximum token count for each page of the documents to estimate chunk size, was used to set a starting point for experimenting. The Python script for calculating token for documents is as below:

Listing 1. Script used to calculate max and average token count for documents used in **rag** embedding

```
from langchain_community.document_loaders
    import PyPDFLoader
import tiktoken

loader = PyPDFLoader(FILEPATH)
pages = loader.load()

tokenizer = tiktoken.get_encoding('
    cl100k_base')
def tiktoken_len(text):
    tokens = tokenizer.encode(
    text,
    disallowed_special=()
)
    return len(tokens)

tiktoken.encoding_for_model('gpt-3.5-
    turbo')

def get_token_count():
    token_counts = []
    for page in pages:
        token_counts.append(tiktoken_len(
            page.page_content))
    min_token_count = min(token_counts)
    avg_token_count = int(sum(
        token_counts) / len(token_counts))
    max_token_count = max(token_counts)

    return min_token_count,
        avg_token_count, max_token_count
```

[2]https://learn.microsoft.com/en-us/azure/search/vector-search-how-to-chunk-documents

The list of included articles are:

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3) J. Melegati and X. Wang, *Digital Mentor: towards a conversational bot to identify hypotheses for software startups*, Apr. 2023.
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7) F. Fagerholm, A. Sanchez Guinea, H. Mäenpää, og J. Münch, *The RIGHT model for Continuous Experimentation*, Journal of Systems and Software, 2017.
8) T. Harris, *Start-up: A Practical Guide to Starting and Running a New Business*, Springer Publishing, 2006.
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how-not-to-fail/.
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14) P. Graham, *Do things that don't scale*, en, https://paulgraham.com/ds.html.
15) M. Seibel, *Fundraising rounds are not milestones*, en, https://www.michaelseibel.com/blog/fundraising-rounds-are-not-milestones.
16) M. Seibel, *The real product-market fit*, en, https://www.ycombinator.com/library/5z- the- real-product- market- fit, Accessed: 2024-04-11
17) S. Blank and B. Dorf, *The Startup Owner's Manual: The Step-by-Step Guide for Building a Great Company*, KS Ranch Publishing Division, 2012.
18) M. Seibel, *User you don't want*, en, https://www.ycombinator.com/library/67-user-you-don-t-want.
19) A. Harris, *Taking advice*, en, https://blog.aaronkharris.com/taking-advice.
20) J. Bosch, H. Olsson, J. Björk and J. Ljungblad, *The early stage software startup development model: A framework for operationalizing lean principles in software startups*, Jan. 2016

III. Research Methodology

The study adopts the Design Science Research Methodology (DSRM), which aims to obtain research insights by developing and evaluating artefacts [11]. We conducted three design science cycles for the development of StartupGPT. In Cycle 1, we explored the most feasible use scenarios of LLMs in consulting software startups. Five scenarios are selected for testing StartupGPT, which are: (1) prototyping and MVP development, (2) product development guidance, (3) assisting in analyzing market, customers and competitors, (4) funding and investment guidance and (5) creating business plan. We also explored the use of prompt engineering in providing effective support to software startups. In Cycle 2, we explored the use of system instructions, a set of fixed instructions created by the developers to constrain the LLM's response. In Cycle 3, we use RAG with a database of empirical knowledge from software startup research. In this paper, we reported the results from Cycle 3.

The prototype in Cycle 3 was tested on a total of 11 users. Users were gathered from **nse**, Gründerbrakka, and through contacts of the main reseacher. To gather users, posters and flyers encouraging users to partake in the user test were handed out and hung up around the **nse** areas at the **ntnu** campus. An overview of the participants in the user test can be seen in Table I.

User tests were conducted online without supervision, through a video chat with supervision, or in the **ux**-lab with supervision. The structure of directly observed tests was consistent, involving filming both the users and their screens as they conducted the test. The observed user tests involved a facilitator explaining the task, followed by the users performing it independently without guidance. The user tests followed a within-subject design, where each user is exposed to more than one of the variations being tested of the artefact [12]. Participants were encouraged to think aloud and share their thoughts throughout the test.

A short semi-structured interview was conducted at the end to further explore the participant's perception of the chatbots and their usability. The researchers follow an interview script for guidance to ensure all topics are covered, but may ask supplementary questions or change the order [13], [14]. This allows the researchers to gather more detailed information about the topics and the interviewees get the chance to introduce issues they find relevant to the questions.

The quantitative data is gathered through a questionnaire. The ordinal Likert data was visualised in stacked bar charts to highlight patterns regarding the ratings across the different gpt models. The qualitative analysis was done by extracting themes and topics from the user tests, including the written comments that were given in the prototype questionnaire and the verbal comments given during the in-person user tests. The recordings were transcribed and analysed to obtain reoccurring themes and quotes, as proposed by [13]. The data was analysed using an inductive approach to identify and categorise meaningful themes and codes from the transcriptions relevant to the research questions [14]. This analysis used the online whiteboard tool Miro.

IV. Result

This section presents the results from the third cycle, obtained through qualitative and quantitative analysis of the feedback gathered during the user tests. Figure 3 presents the mean value of user evaluation on the four dimensions: effective, efficient, reliable and satisfactory. The chart compares three types of interventions for LLM outcomes: LLM-based chatbot integrating demographic information (red), LLM-based chatbot integrating both demographic information and RAG (yellow) and LLM-based chatbot integrating previous treaments and system instructions. The figure shows that ovreall the user is positive towards these chatbots, however, the effect of embedding research articles via RAG is not significant.

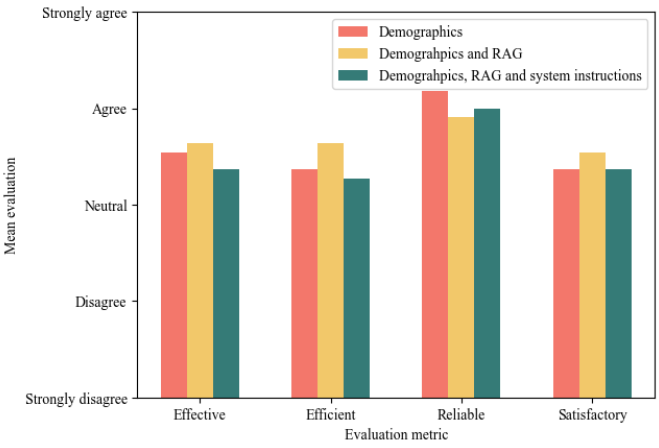


Fig. 3. Mean user evaluations for the chatbots

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User	Company Role	Stage	Industry	Year of business	ChatGPT experience	Year of birth	Location	Type of user test
User 1	Chairman	Seed	Manufacturing	1-5 years	Beginner	50-70 years	Malmö, Sweden	Direct observation
User 2	CFO	Seed	Information and Communication	1-5 years	Experienced	18-30 years	Trondheim, Norway	Direct observation
User 3	CEO	Early	Information and Communication	1-5 years	Experienced	18-30 years	Trondheim, Norway	Direct observation
User 4	CPO	Seed	Transportation and Storage	1-5 years	Beginner	18-30 years	Trondheim, Norway	Direct observation
User 5	CEO	Seed	Professional, Scientific and Technical Activities	¡1 year	Intermediate	50-70 years	Oslo, Norway	Direct observation
User 6	CEO	Seed	Human Health and Social Work Activities	¡1 year	Intermediate	18-30 years	Trondheim, Norway	Indirect observation
User 16	CTO	Seed	Administrative and Support Service Activities	¡1 year	Experienced	18-30 years	Trondheim, Norway	Direct observation
User 17	CEO	Seed	Professional, Scientific and Technical Activities	¡1 year	Experienced	18-30 years	Oslo, Norway	Direct observation
User 18	Founder	Expansion	Professional, Scientific and Technical Activities	¿10 year	No experience	50-70 years	Bergen, Norway	Direct observation
User 19	Developer	Early	Information and Communication	1-5 years	Experienced	18-30 years	Trondheim, Norway	Indirect observation
User 20	CTO	Early	Information and Communication	1-5 years	Intermediate	Undisclosed	Uruguay	Indirect observation

TABLE I
OVERVIEW OF THE PARTICIPANTS IN THE USER TESTS FOR CYCLE 3

Insight from user interviews provide details on the positive and negative sides of RAG for StartupGPT:

A. Positive feedbacks

Overall participants appreciate the usefulness, reliability and relevance of RAG-based chatbot, in comparing to normal GPT models.

1) Increased Usefulness: Some users commented that they appreciated the framework related advice they received. User 3 found the answers from the chatbots using the **rag** model to be more startup relevant and User 16 found several of the framework related suggestions from the chatbot using demographics and **rag** to be useful, such as the **bml** loop.

> *"That one is key [about the **bml** loop]. I haven't heard of it before, but it's definitely very, very good."* (User 16)

2) Increased Reliability: Users 3, 4, 5, 16, 18, and 19 all stated that they perceived the answers from all chatbots to be reliable.

> *"I actually trusted all of them [...] They all gave good answers. I agreed with everything they said."* (User 16)

One of the main advantages of including RAG in the chatbots was to further ensure reliability of the answers and reduce the chance of hallucinations. In Cycle 2, User 3 experienced that the chatbot suggested the **esssdm** framework, but defined it incorrectly. When the chatbots wrote about framework-related theory in this cycle, for instance suggesting the **bml** loop for User 16, they were defined correctly according to the articles

3) Increased Relevance: However, the users were positive when they were met with concrete examples. User 1 commented that the chatbots in general were *"to the point"*, and User 3 was positive to examples of real companies, provided by the chatbot using demographics and **rag**, arguing that it made this chatbot the most specific.

> *"The fact that it provided real examples was very good [...] then I can faster dive into those cases and enhances my learning process"* (User 3)

In contrary to the responses from Cycle 2, none of the users experience hallucinations during their conversations and the need for references was not mentioned. While this could be by chance, it could be an indication that the users found the answers more trustworthy and did not feel the same need for assurance that the answers were reliable.

However, the implementation of more quality assured information through **rag** only applies to information that can be found in the knowledge base, thus, some users still experienced some errors. User 1 experienced that the chatbot using demographics and **rag** made an assumption about the product that would have been impossible.

B. Negative feedbacks

As room for improvement, the current version of StartupGPT lack of sufficient concrete details, interactive conversation style, too theoretical and lack of personality

1) Lack of concrete details: User 5 commented that the advice they received did not exceed general recommendations for startups.

> *"If you have had one course at BI [Norwegian Business School], you would already know these answers"* (User 5)

User 2 and 17 commented that the answers were correct, but they wanted more concrete examples, while user 16 wanted the chatbot to take a more guiding role and provide them with more detailed responses.

> *"I think they are relatively fine, maybe a bit vague, maybe a bit safe. It's decent information that I could use to ask further questions."* (User 2)
>
> *"This is nice and I agree, but what does is give me? Give me something concrete"* (User 17)

Several users pointed out that this would help provide more specific and relevant answers for each company. This feedback was not aimed more towards one chatbot, but rather general feedback from the users.

> *"It would be nice if it could incorporate even more company-specific information. The answers sought are often very individual for each business, based*

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on customers, etc. The information also changes constantly, so feeding in even more information to get more targeted answers would be beneficial." (User 4)

2) Interactive conversation style: Users 2, 3 and 16 suggested that the chatbot could ask, or "prompt", the user back to obtain relevant information and build context, before the user starts asking questions.

> *"I would have preferred to early on see prompts [aimed at the user] requesting more information to solidify the context, like: 'I see that you're asking about this, but to explore this path effectively, I need to know your thoughts on these aspects first.' "* (User 2)
>
> *"I really want it to prompt me first. Like, 'Where are you now? MVP? Testing? Marketing? etc...' [Then I can respond with] 'We are in MVP, we have a pilot.' [And then it could ask] 'Okay, how are you working with loop learning?" [...]'. So I can answer that, and then: 'Here are some alternatives you should check out.' [...]*
>
> *You don't always know what to ask. So it's nice to be prompted into identifying which phase you are in."* (User 3)
>
> *"What I feel is missing from all three is that none of them ask me more about my idea. They are only basing their responses on a single sentence about what I intend to do. [...] Either it should ask me more to understand the context of my idea, or I should provide a very detailed description of my idea from the start, so it has that information."* (User 16)

User 17 also suggested that the chatbot could gain context by providing the user with a set of statements about their startup for them to take a stance on.

> *"[I would prefer it to] map out my current status more thoroughly. That I could take a stance on various statements and make choices continuously, maybe with brief descriptions as well."* (User 17)

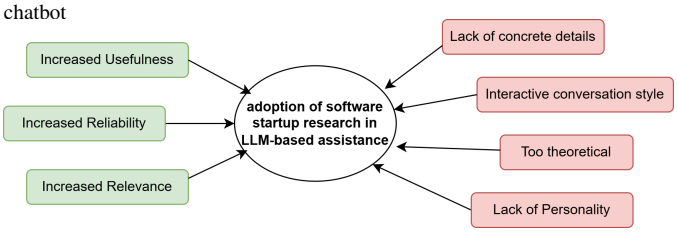
3) Too theoretical:: Increased focus on framework theory may have made the responses too academic, as Users 4, 5, and 18 responded that the answers were too theoretical and lacked practical, applicable advice.

> *"This was just a schoolbook summary, which is mostly impractical, not relevant, and may just create more thresholds. I found it to be too theoretically complicated."* (User 18)

User 18 further explained that startup frameworks could be useful for many startups, but they should be presented more practically.

> *"You should find the business idea before you find the business model, but people don't need to know what a business model is. [...] They don't need to know 18 different paradigms for a business model. [...] You just need to know who you are targeting."* (User 18)

Fig. 4. Summary of user feedback from testing the research-embedded LLM chatbot



V. Discussions and Conclusions

A. Answering the Research Question

We discussed several insights that response to our RQ: "How software startup research outcome can be integrated in LLMs to assist software startups in business and product development?" Response from StartupGPT was often seen as theoretical. Although the users recognised the guidelines as relevant startup frameworks, they found them too standardised to be applicable for their individual startups. There was a general consensus amongt the users that they appreciated practical examples. Considering the startups did not like how the answer seemed too standardised, a more creative approach by the chatbots may to a larger degree live up to the users' expectations. This corresponds with the general consensus of ChatGPT users, that ChatGPT often tends to struggle with creating creative responses and providing practical examples [15].

One of the most prominent feedback points was that the chatbot should gain a deeper understanding of the startup to provide more context-specific answers. The benefits of giving the chatbot a clear context and descriptions of the users can also be supported by other studies. While there has been done little research on this in a startup context, several studies have been done in other segments. For instance, Belda-Medina and Kokošková examined the role of chatbots as tutors in education and discovered that satisfaction with chatbots decreased when they couldn't tailor their responses to individual students knowledge, needs, and preferences [16]. Furthermore, a study exploring context-aware conversational chatbots found the quality of responses improved due to the context helping the system gain an understanding of the user intent, resulting in more accurate and relevant responses [17].

Feedback from the users suggests that the selection of RAG materials in the knowledge base could have contributed to the lack of perceived usefulness. The current knowledge base for the rag model consists of both research articles and blog posts by leading entrepreneurs, aiming to provide content with both academic and practical guidance. Considering the user feedback, the knowledge base may still have been too focused on research articles and books, lacking shorter explanations and practical examples of applications. A more practically aimed knowledge base could also improve the relevance of the responses. Increased focus on information and real-life examples regarding practical implementation of startup

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frameworks and guidance could, therefore, also be a way to increase the perceived relevance and usability.

The modest appreciation of theoretical frameworks introduced via StartupGPTs may also be due to a general lack of interest in adapting to startup frameworks among entrepreneurs. The adoption of startup frameworks is low. Entrepreneurs tend to focus on business and product development, and adjusting workflows to align with theoretical frameworks is not often prioritised. While the exact reason for this is not clearly defined, Tegegne et al. argue that lack of knowledge and experience from owners and limited resources are key reasons for this low adoption [18]. Presenting the information obtained from the rag model in a way that increases the knowledge of the frameworks and motivates the entrepreneurs to apply them in their startup could therefore be viewed as essential for increasing the framework adoption among startups. As startups are particularly constrained regarding time and resources, it is likely the motivation would need to stem from being shown how using these frameworks would benefit their particular startup.

Furthermore, efficient application of the theoretical frameworks would involve applying the knowledge from the frameworks to give actionable advice specific to the individual entrepreneur. Based on this, the rag-model would thereby be the knowledge base ensuring correctness and extensive knowledge about effective startup methods. However, the answers created by the llm should aim to apply and target this knowledge to offer practical and actionable advice to the particular startup.

B. Limitations

This study's research heavily relies on qualitative analysis to capture a broader range of user experiences than quantitative methods alone could provide. Since the researchers conducted each stage of the research themselves, maintaining complete objectivity in analyzing the results might be subject to bias. Interpretation of user test responses, identifying themes and codes, and determining the main conclusions may have been somewhat influenced by the researchers' biases and expectations.

Most participants in the user tests were recruited through NSE, Gründerbrakka, or the researchers' networks, meaning they likely shared similar backgrounds. The majority were NTNU students in Trondheim, Norway, aged 18-30 and often held C-level roles, potentially limiting their representativeness of a typical entrepreneur. Given that many participants were either current or former NSE students or had extensive experience with startups, they likely possessed a higher level of knowledge than one might expect from an average person in an early-stage startup. This background could have influenced their perception of the chatbot's value.

C. Related work

Several startup frameworks are embedded into StartupGPT, including the build-measure-learn loop [19], RIGHT model for continuous experimentation [20], esssdm [21], HyMap [22], 6W3H [23] and StartCards [24]. Several previous researches have shown software startups' adoption of startup frameworks, such as Lean Startup, remains low [21], [25], [26]. Although the adaptation of frameworks is recommended to increase a startup's probability of succeeding [21], a study by Pantiuchina et al. found out that only 15 % of the startups they studied followed Lean Startup [26]. Interviews conducted by Bosch et al. show that many startups find it difficult to implement and apply the frameworks in practice [21]. Furthermore, Bosch et al. explained that in a startup that is still finding both the problem and solution of their product, agile methods seem less relevant as they typically rely on a well-defined problem [21].

The possibilities for the development of a digital mentor aimed at aiding startups have been researched by Melegati and Wang [22], [27]. Their work focuses on the development of hypotheses for startup experimenting, which is a common technique used to reduce uncertainty in startups. Their previous work includes the development of the HyMap technique: a technique based on mapping the entrepreneurs' understanding of their market and customers to identify the hypotheses. The HyMap technique relies on a facilitator to interview the startups, but their proposed solution for the Digital Mentor replaces the human facilitator with a conversational bot, to make the resource more available. While this approach is still in ideation phase, we believe that they can learn from experience with StartupGPT.

D. Future work

Findings from the research process uncovered several areas for improvement, that could be used as a basis for further development and research. Feedback from the users shows that measures to increase the specificity and relevance of the responses and adjusting the chatbot to gain information from the user, would increase the usability.

Further, the field of generative ai and ChatGPT is in constant development and new possibilities have opened up, also over the duration of this study. Experimentation forward could focus on further exploring different ways to customise ChatGPT and the rag model. Other technologies used with generative ai, such as fine-tuning and iterative refinement with self-feedback, could be interesting to explore.

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